Fractions

Addition, Subtraction, Multiplication and Division

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Churchill College | Mathematics

Session Flow: From Shapes to Operations

A 100-minute student-centered lesson structured on Buckingham's PGC Framework.

Lesson Roadmap (100 min)

- **10 min – Phase 1:** Novelty & Curiosity (The Infinite Square Area Puzzle)
- **25 min – Phase 2:** Expectation & Motivation (Rule Deduction: Addition, Multiplication & Division)
- **40 min – Phase 3:** Effort & Scaffolding (Level 1, Level 2, Level 3 Practices & Independent Work)
- **25 min – Phase 4:** Recognition & Peer Feedback (Co-evaluation & Synthesis)

1. Novelty & Curiosity | The Infinite Square Puzzle

Driving Question

"If we take a square of Area $= 1$ and repeatedly cut the remaining piece in half forever... what is the area of each slice, and what happens if we add them all up?"

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Think-Pair-Share (3 min)

- Why does each cut create a unit fraction?
- What is the sum of $\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \ldots$?
- Discuss with your partner and record your hypothesis.

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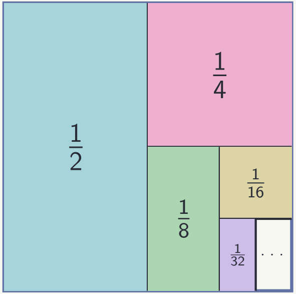

2. Expectation & Motivation | Deducing Addition Rules

Guided Discovery (Pair Discussion - 4 min): Observe the patterns to derive the structural rules.

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Case A: Same Denominator

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Case B: Integer + Fraction

$$3 + \frac{2}{5} = \frac{3}{1} + \frac{2}{5} = \frac{15+2}{5} = \frac{17}{5}$$

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Case C: Different Denominators

$$\frac{1}{3} + \frac{2}{5} = \frac{1 \times 5 + 2 \times 3}{3 \times 5} = \frac{5+6}{15} = \frac{11}{15}$$

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Rule: Find common denominator (cross-multiply or LCM).

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Case D: Multi-Fraction ($>$ 3 Terms)

Use the **Least Common Multiple (LCM)** of all denominators to combine in a single step!

Spot the Pattern (2 min): Unlike addition, fraction multiplication does **NOT** require a common denominator!

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Fraction $\times$ Fraction

Multiply straight across (numerator $\times$ numerator, denominator $\times$ denominator):

$$\frac{a}{b} \times \frac{c}{d} = \frac{a \times c}{b \times d}$$

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Fraction × Fraction

Multiply straight across (numerator × numerator, denominator × denominator):

$$\frac{a}{b} \times \frac{c}{d} = \frac{a \times c}{b \times d}$$

Example:

$$\frac{3}{4} \times \frac{5}{7} = \frac{3 \times 5}{4 \times 7} = \frac{15}{28}$$

Integer × Fraction

Complete the integer with a denominator of 1, then multiply straight across:

$$n \times \frac{a}{b} = \frac{n}{1} \times \frac{a}{b} = \frac{n \times a}{1 \times b}$$

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$$n \times \frac{a}{b} = \frac{n}{1} \times \frac{a}{b} = \frac{n \times a}{1 \times b}$$

Example:

$$4 \times \frac{2}{9} = \frac{4}{1} \times \frac{2}{9} = \frac{8}{9}$$

2. Expectation & Motivation | Deducing Division Rules (Part 1)

Spot the Pattern (3 min): Division in fractions is converted directly into multiplication using reciprocals!

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Method 1: Keep – Switch – Flip (KSF)

To divide $\frac{a}{b} \div \frac{c}{d}$:

1. **Keep** the 1st fraction: $\frac{a}{b}$
2. **Switch** $\div$ to $\times$
3. **Flip** the 2nd fraction: $\frac{d}{c}$

$$\frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \times \frac{d}{c} = \frac{a \times d}{b \times c}$$

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Method 2: Butterfly Multiplication

Cross-multiply directly in a diagonal pattern:

- **Top-Left** $\times$ **Bottom-Right**
- **Bottom-Left** $\times$ **Top-Right**

$$\frac{a}{b} \div \frac{c}{d} = \frac{a \times d}{b \times c}$$

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$$\frac{a}{b} \div \frac{c}{d} = \frac{a \times d}{b \times c}$$

Example:

$$\frac{2}{3} \div \frac{5}{7} = \frac{2 \times 7}{3 \times 5} = \frac{14}{15}$$

2. Expectation & Motivation | Deducing Division Rules (Part 2)

Complex Fractions: When a fraction is divided by another fraction formatted vertically.

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The Sandwich Law

For a complex fraction $\frac{\frac{a}{b}}{\frac{c}{d}}$:

- **Bread (Extremes):** Multiply top and bottom numbers → **Numerator** $(a \times d)$
- **Filling (Means):** Multiply inner numbers → **Denominator** $(b \times c)$

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Step-by-Step Example

Compute:

$$\frac{\frac{3}{4}}{\frac{5}{7}}$$

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Step-by-Step Example

Compute:

$$\frac{\frac{3}{4}}{\frac{5}{7}}$$

- **Extremes (Bread):** $3 \times 7 = 21$
- **Means (Filling):** $4 \times 5 = 20$

$$\frac{\frac{3}{4}}{\frac{5}{7}} = \frac{3 \times 7}{4 \times 5} = \frac{21}{20}$$

[Level 1: Easy – Individual Work | 5 min]

Goal: Direct application to build immediate confidence and earn early dopamine wins.

Solve the following in your notebook:
Part A: Addition & Multiplication

1. $\frac{4}{11} + \frac{5}{11} =$

[Level 1: Easy – Individual Work | 5 min]

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Solve the following in your notebook:
Part A: Addition & Multiplication

1. $\frac{4}{11} + \frac{5}{11} = \frac{9}{11}$
2. $2 + \frac{3}{7} =$

3. Effort & Scaffolding | Level 1: Direct Application

[Level 1: Easy – Individual Work | 5 min]

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Solve the following in your notebook:

Part A: Addition & Multiplication

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2. $2 + \frac{3}{7} = \frac{2}{1} + \frac{3}{7} = \frac{14+3}{7} = \frac{17}{7}$

Part B: Multiplication

1. $5 \times \frac{3}{16} =$

[Level 1: Easy – Individual Work | 5 min]

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Part A: Addition & Multiplication

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Part B: Multiplication

1. $5 \times \frac{3}{16} = \frac{5}{1} \times \frac{3}{16} = \frac{15}{16}$
2. $\frac{2}{5} \times \frac{4}{7} =$

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Part B: Multiplication

1. $5 \times \frac{3}{16} = \frac{5}{1} \times \frac{3}{16} = \frac{15}{16}$
2. $\frac{2}{5} \times \frac{4}{7} = \frac{2 \times 4}{5 \times 7} = \frac{8}{35}$

3. Effort & Scaffolding | Level 1: Division Application

[Level 1: Easy – Individual Work | 5 min]

Goal: Direct application of division rules to reinforce standard techniques.

Solve the following in your notebook:
Part A: Sandwich Law (Complex Fractions)

1. $\frac{\frac{1}{3}}{\frac{2}{5}} =$

[Level 1: Easy – Individual Work | 5 min]

Goal: Direct application of division rules to reinforce standard techniques.

Solve the following in your notebook:
Part A: Sandwich Law (Complex Fractions)

1. $\dfrac{\frac{1}{3}}{\frac{2}{5}} = \dfrac{1 \times 5}{3 \times 2} = \dfrac{5}{6}$
2. $\dfrac{3}{\frac{4}{5}} =$

3. Effort & Scaffolding | Level 1: Division Application

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Solve the following in your notebook:

Part A: Sandwich Law (Complex Fractions)

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2. $\frac{3}{\frac{4}{5}} = \frac{\frac{3}{1}}{\frac{4}{5}} = \frac{3 \times 5}{1 \times 4} = \frac{15}{4}$

Part B: Division (KSF & Butterfly)

1. $\frac{3}{4} \div \frac{2}{5} =$

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Part B: Division (KSF & Butterfly)

1. $\frac{3}{4} \div \frac{2}{5} = \frac{3}{4} \times \frac{5}{2} = \frac{15}{8}$
2. $4 \div \frac{2}{7} =$

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Part B: Division (KSF & Butterfly)

1. $\frac{3}{4} \div \frac{2}{5} = \frac{3}{4} \times \frac{5}{2} = \frac{15}{8}$

2. $4 \div \frac{2}{7} = \frac{4}{1} \times \frac{7}{2} = \frac{28}{2} = 14$

3. Effort & Scaffolding | Level 2: Spot the Mistake & LCM

[Level 2: Medium – Pair Work | 7 min]

Goal: Error detection and multi-denominator addition.

Task A: Spot the Misconception!

A student wrote on the board:

$$\frac{2}{5} + \frac{1}{3} = \frac{3}{8} \quad \textbf{(WRONG!)}$$

Your Task: Identify the error and solve it correctly using a common denominator.

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Correction: Never add denominators!
$\frac{2}{5} + \frac{1}{3} = \frac{6+5}{15} = \frac{11}{15}$.

Task B: Different Denominators

Calculate:

$$\frac{3}{8} + \frac{5}{6}$$

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Task B: Different Denominators

Calculate:

$$\frac{3}{8} + \frac{5}{6}$$

$\mathrm{LCM}(8, 6) = 24$.

$$\frac{3 \times 3}{24} + \frac{5 \times 4}{24} = \frac{9 + 20}{24} = \frac{29}{24}$$

3. Effort & Scaffolding | Level 2: Division Misconceptions

[Level 2: Medium – Pair Work | 7 min]

Goal: Error detection, multi-denominator addition, and complex fraction division.

Task A: Spot the Misconception!

A student wrote on the board:

$$\frac{2}{5} \div \frac{3}{4} = \frac{2 \div 3}{5 \div 4} \quad \textbf{(WRONG!)}$$

Your Task: Identify the error and solve correctly using Keep–Switch–Flip.

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$\frac{2}{5} \times \frac{4}{3} = \frac{8}{15}$.

Task B: Sandwich Law Application

Calculate and simplify:

$$\frac{\frac{5}{6}}{\frac{2}{3}}$$

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Your Task: Identify the error and solve correctly using Keep–Switch–Flip.

Correction: Flip the 2nd fraction! $\frac{2}{5} \times \frac{4}{3} = \frac{8}{15}$.

Task B: Sandwich Law Application

Calculate and simplify:

$$\frac{\frac{5}{6}}{\frac{2}{3}}$$

Extremes: $5 \times 3 = 15$

Means: $6 \times 2 = 12$

$$\frac{\frac{5}{6}}{\frac{2}{3}} = \frac{15}{12} = \frac{5}{4}$$

3. Effort & Scaffolding | Level 3: Multi-Fraction LCM Challenge

[Level 3: Hard – Small Group Challenge | 8 min]

Goal: High-level reasoning and multi-term LCM synthesis.

Problem 1 (LCM of $>$3 fractions): Compute the single simplified fraction:

$$\frac{1}{2}+\frac{2}{3}+\frac{3}{4}+\frac{1}{6}$$

[Level 3: Hard – Small Group Challenge | 8 min]

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Step-by-Step LCM Solution

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Step-by-Step LCM Solution

1. Find $\mathrm{LCM}(2, 3, 4, 6) = 12$.

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Step-by-Step LCM Solution

1. Find $\mathrm{LCM}(2,3,4,6)=12$.
2. Convert each term: $\frac{1}{2}=\frac{6}{12}$, $\quad\frac{2}{3}=\frac{8}{12}$, $\quad\frac{3}{4}=\frac{9}{12}$, $\quad\frac{1}{6}=\frac{2}{12}$.

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Step-by-Step LCM Solution

1. Find $\mathrm{LCM}(2, 3, 4, 6) = 12$.
2. Convert each term: $\frac{1}{2} = \frac{6}{12}, \quad \frac{2}{3} = \frac{8}{12}, \quad \frac{3}{4} = \frac{9}{12}, \quad \frac{1}{6} = \frac{2}{12}$.
3. Combine: $\frac{6+8+9+2}{12} = \frac{25}{12}$.

3. Effort & Scaffolding | Level 3: Multi-Operation Challenge

[Level 3: Hard – Small Group Challenge | 8 min]

Goal: High-level reasoning, multi-term LCM, and nested operations synthesis.

Problem 1 (Complex Mixed Operations): Simplify the following fraction:

$$\frac{\frac{1}{2}+\frac{1}{3}}{\frac{5}{6}}$$

[Level 3: Hard – Small Group Challenge | 8 min]

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Step-by-Step Solution

1. **Numerator Addition:** $\frac{1}{2}+\frac{1}{3}=\frac{3+2}{6}=\frac{5}{6}$.

3. Effort & Scaffolding | Level 3: Multi-Operation Challenge

[Level 3: Hard – Small Group Challenge | 8 min]

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Problem 1 (Complex Mixed Operations): Simplify the following fraction:

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Step-by-Step Solution

1. **Numerator Addition:** $\frac{1}{2}+\frac{1}{3}=\frac{3+2}{6}=\frac{5}{6}$.
2. **Apply Sandwich Law:** $\frac{\frac{5}{6}}{\frac{5}{6}}$.

[Level 3: Hard – Small Group Challenge | 8 min]

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Problem 1 (Complex Mixed Operations): Simplify the following fraction:

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Step-by-Step Solution

1. **Numerator Addition:** $\frac{1}{2}+\frac{1}{3}=\frac{3+2}{6}=\frac{5}{6}$.
2. **Apply Sandwich Law:** $\frac{\frac{5}{6}}{\frac{5}{6}}$.
3. **Extremes / Means:** $\frac{5\times 6}{6\times 5}=\frac{30}{30}=1$.

4. Recognition & Feedback | Peer Co-Evaluation Protocol

Notebook Exchange (8 min)

Co-Evaluation Protocol

1. **Generative Challenge:** Create 2 fraction exercises in your notebook for a peer to solve.
2. **Peer Review:** Swap notebooks and solve the exercises. Then review their solution.

1. **Praise (Top Bread):** Highlight a clear step or correct usage of the rules.
2. **Constructive Insight (Filling):** Point out any arithmetic mistake.
3. **Validation (Bottom Bread):** Sign your peer's page with an encouraging summary note.

Classroom Submission

Digital Exit Ticket: Take a photo of your peer-evaluated sheet and upload it to **Google Classroom** before leaving!

Operation Type	Key Procedural Strategy
Integer + Fraction	

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Integer $+$ Fraction	Complete integer as $\frac{n}{1}$ before adding.

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Integer + Fraction	Complete integer as $\frac{n}{1}$ before adding.
Same Denominator Addition	

Operation Type	Key Procedural Strategy
Integer + Fraction	Complete integer as $\frac{n}{1}$ before adding.
Same Denominator Addition	Add numerators directly, keep denominator.

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Integer + Fraction	Complete integer as $\frac{n}{1}$ before adding.
Same Denominator Addition	Add numerators directly, keep denominator.
Different Denominators	

Operation Type	Key Procedural Strategy
Integer + Fraction	Complete integer as $\frac{n}{1}$ before adding.
Same Denominator Addition	Add numerators directly, keep denominator.
Different Denominators	Convert using cross-multiplication or LCM.

Operation Type	Key Procedural Strategy
Integer + Fraction	Complete integer as $\frac{n}{1}$ before adding.
Same Denominator Addition	Add numerators directly, keep denominator.
Different Denominators	Convert using cross-multiplication or LCM.
>3 Fractions Addition	

4. Recognition & Feedback | Session Synthesis

Operation Type	Key Procedural Strategy
Integer $+$ Fraction	Complete integer as $\frac{n}{1}$ before adding.
Same Denominator Addition	Add numerators directly, keep denominator.
Different Denominators	Convert using cross-multiplication or LCM.
> 3 Fractions Addition	Find global LCM of all denominators.

4. Recognition & Feedback | Session Synthesis

Operation Type	Key Procedural Strategy
Integer $+$ Fraction	Complete integer as $\frac{n}{1}$ before adding.
Same Denominator Addition	Add numerators directly, keep denominator.
Different Denominators	Convert using cross-multiplication or LCM.
>3 Fractions Addition	Find global LCM of all denominators.
Fraction $\times$ Fraction	

Operation Type	Key Procedural Strategy
Integer $+$ Fraction	Complete integer as $\frac{n}{1}$ before adding.
Same Denominator Addition	Add numerators directly, keep denominator.
Different Denominators	Convert using cross-multiplication or LCM.
>3 Fractions Addition	Find global LCM of all denominators.
Fraction $\times$ Fraction	Multiply numerators together and denominators together.

Operation Type	Key Procedural Strategy
Integer $+$ Fraction	Complete integer as $\frac{n}{1}$ before adding.
Same Denominator Addition	Add numerators directly, keep denominator.
Different Denominators	Convert using cross-multiplication or LCM.
>3 Fractions Addition	Find global LCM of all denominators.
Fraction $\times$ Fraction	Multiply numerators together and denominators together.
Fraction $\div$ Fraction (KSF)	

Operation Type	Key Procedural Strategy
Integer $+$ Fraction	Complete integer as $\frac{n}{1}$ before adding.
Same Denominator Addition	Add numerators directly, keep denominator.
Different Denominators	Convert using cross-multiplication or LCM.
> 3 Fractions Addition	Find global LCM of all denominators.
Fraction $\times$ Fraction	Multiply numerators together and denominators together.
Fraction $\div$ Fraction (KSF)	Keep 1st, Switch $\div \to \times$, Flip 2nd $(\frac{a}{b} \times \frac{d}{c})$.

Operation Type	Key Procedural Strategy
Integer $+$ Fraction	Complete integer as $\frac{n}{1}$ before adding.
Same Denominator Addition	Add numerators directly, keep denominator.
Different Denominators	Convert using cross-multiplication or LCM.
>3 Fractions Addition	Find global LCM of all denominators.
Fraction $\times$ Fraction	Multiply numerators together and denominators together.
Fraction $\div$ Fraction (KSF)	Keep 1st, Switch $\div \to \times$, Flip 2nd ($\frac{a}{b} \times \frac{d}{c}$).
Complex Fraction (Sandwich Law)	

4. Recognition & Feedback | Session Synthesis

Operation Type	Key Procedural Strategy
Integer + Fraction	Complete integer as $\frac{n}{1}$ before adding.
Same Denominator Addition	Add numerators directly, keep denominator.
Different Denominators	Convert using cross-multiplication or LCM.
>3 Fractions Addition	Find global LCM of all denominators.
Fraction $\times$ Fraction	Multiply numerators together and denominators together.
Fraction $\div$ Fraction (KSF)	Keep 1st, Switch $\div \to \times$, Flip 2nd ($\frac{a}{b} \times \frac{d}{c}$).
Complex Fraction (Sandwich Law)	Bread (Extremes) over Filling (Means) $\to \frac{a \cdot d}{b \cdot c}$.

Fantastic work constructing fraction fluency today!

50 Minutes of Focused Individual Work

Task Objectives & Instructions

- **Individual Focus:** Work independently on your assigned task/homework.
- **Self-Paced Progression:** Move sequentially through the problem sets or task guidelines.
- **Resource Management:** Consult your reference notes, class slides, or guidelines before asking questions.

Expectations

- Silent and continuous effort.
- Complete all required steps in your notebook/file.

Teacher Support

- Raise your hand for targeted 1-on-1 feedback.
- Check progress against the success criteria.

Timer: 50:00 min – Maximize your active working time!